OPERATIONAL SUSTAINABILITY

Resource Management

Energy Efficiency · Water Conservation · Waste Reduction · Circular Economy Principles

The global economy is only 6.9% circular. The other 93% is resources used once, then lost.

Today is entirely practical: the actual calculations engineers and facility managers use to cut energy, water, and waste — and the systems-level thinking that ties them together.

TODAY'S JOURNEY

Four Resources, One Systems Problem



01

Energy Efficiency

Doing the same work with less power — and calculating whether it pays for itself.



02

Water Conservation

Protecting the resource India is running out of fastest.



03

Waste Reduction

The hierarchy of what to do with material before it becomes garbage.



04

The Circular Economy

The systems-level model that ties all three together permanently.

SECTION 1 OF 4

Energy Efficiency

Delivering the same output, service, or comfort — while using measurably less power.

SECTION 1 · DEFINITION

What Is Energy Efficiency?

A precise engineering concept, often confused with its cousin, energy conservation.

The Working Definition

Using less energy to provide the same product, service, or level of comfort — doing the same job, with less input.

Efficiency vs. Conservation

Conservation reduces energy use by reducing the service itself (turning off a light); efficiency reduces energy use while keeping the service the same (a more efficient light, on for the same hours).

Energy Intensity: The Macro View

Energy used per unit of economic output (e.g., MJ per unit of GDP) — a country-level indicator of how efficiently an economy converts energy into value.

SECTION 1 · WHERE IT PAYS OFF

Four Levers of Energy Efficiency

The same underlying logic — do more with less — applied across every major energy-consuming system.



Buildings & HVAC

Insulation, smart thermostats, and efficient heating/cooling — buildings account for a large share of global energy demand.



Industrial Processes

Waste heat recovery, efficient motors, and process optimization — often the fastest payback in an industrial facility.



Lighting

LED lighting uses a small fraction of the energy of incandescent bulbs for the same brightness — one of the simplest, fastest wins available.



Transport & Logistics

Fuel efficiency standards, route optimization, and electrification reduce energy use per kilometre travelled.

SECTION 1 · MAKING IT SYSTEMATIC

ISO 50001: Energy Management Systems

Efficiency gains stick when they're built into a management system — not left to one-off projects.

What It Is

The international standard for establishing, implementing, and continually improving an energy management system within an organization.

The Core Cycle

A Plan-Do-Check-Act cycle: set an energy baseline and targets, implement improvements, measure results, and adjust — repeated continuously.

Why It Matters

It turns "we should save energy" into an auditable, repeatable organizational process — the same logic industrial quality systems use for defects.

SECTION 1 · WORKED EXAMPLE

Payback Period on an LED Lighting Retrofit

GIVEN

- 200 incandescent bulbs (60 W each), operating 10 hours/day
- Replacing with 200 LED bulbs (9 W each)
- Electricity cost: ₹7/kWh; Retrofit cost: ₹250/bulb (₹50,000 total)

FORMULA

Payback Period (years) = Total Investment ÷ Annual Cost Savings

STEP-BY-STEP CALCULATION

1. Power saved per bulb: 60W − 9W = 51W = 0.051 kW
2. Daily savings (200 bulbs): 0.051 kW × 10 hrs × 200 = 102 kWh/day → 37,230 kWh/year
3. Annual cost savings: 37,230 kWh × ₹7/kWh = ₹2,60,610/year
4. Payback: ₹50,000 ÷ ₹2,60,610 ≈ 0.19 years ≈ 2.3 months

≈ 2.3 Months

The retrofit pays for itself in under three months — incandescent-to-LED conversions are among the fastest-payback efficiency investments that exist.

◆ INTERACTIVE

THINK & SHARE

"Of the four levers — buildings, industry, lighting, transport — which do you think offers the fastest payback for a typical Indian college campus, and why?"

Think about what's cheapest to retrofit versus what saves the most energy.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

What is the key difference between energy efficiency and energy conservation?

A They are exactly the same thing

B Efficiency keeps the service the same while using less energy; conservation reduces the service itself ✓

C Conservation only applies to industry, efficiency only to buildings

D Efficiency always costs more than conservation

ANSWER & WHY

B. Efficiency keeps the service the same while using less energy; conservation reduces the service itself

Energy conservation reduces energy use by reducing or eliminating the service (e.g., turning off a light). Energy efficiency reduces energy use while keeping the exact same service level (e.g., a more efficient light, on for the same number of hours) — a subtle but important engineering distinction.

SECTION 2 OF 4

Water Conservation

Protecting the resource that India, specifically, is running out of fastest.

SECTION 2 · THE URGENCY

India's Water Stress, By The Numbers

~600 M

Indians facing high to extreme water stress (NITI Aayog, CWMI)

820 M

People live in river basins with per capita availability at or below the scarcity threshold

1,486 m³

India's per capita water availability (2021), projected to fall toward 1,140 m³ by 2050

18% / 4%

Share of world's population India holds, versus share of world's freshwater resources

India's water demand is projected to be twice its available supply by 2030 — an estimated 6% loss of GDP if unaddressed.

Source: NITI Aayog, Composite Water Management Index (CWMI); Ministry of Jal Shakti.

SECTION 2 · THE TOOLKIT

Four Tools for Water Conservation

Interventions available at every scale, from a single building to an entire watershed.



Low-Flow Fixtures

Efficient taps, showerheads, and dual-flush toilets can cut household and institutional water use significantly with no change in service.



Rainwater Harvesting

Capturing rooftop and surface runoff for direct use or groundwater recharge — cheap, decentralized, and highly effective at scale.



Wastewater Recycling

Treating and reusing greywater and treated sewage for irrigation, cooling, or flushing — reducing demand on freshwater sources.



Precision Irrigation

Drip and micro-irrigation target water directly at crop roots — critical since agriculture consumes roughly 85% of India's water use.

SECTION 2 · WORKED EXAMPLE

Calculating Rainwater Harvesting Potential

GIVEN

- Rooftop catchment area: 500 m^2
- Average annual rainfall in the region: 900 mm (0.9 m)
- Runoff coefficient for a concrete roof: 0.85 (85% collection efficiency)

FORMULA

Harvestable Water (litres) = Area (m^2) × Rainfall (m) × Runoff Coefficient × 1000

STEP-BY-STEP CALCULATION

1. Volume of rain falling on the roof: 500 m^2 × 0.9 m = 450 m^3
2. Apply the runoff coefficient: 450 m^3 × 0.85 = 382.5 m^3
3. Convert to litres: 382.5 × 1,000 = 3,82,500 litres

3.83 Lakh L/Year

This single rooftop can realistically harvest nearly 4 lakh litres of water annually — enough to meaningfully offset a building's freshwater demand.

SECTION 2 · THE NATIONAL RESPONSE

Jal Shakti Abhiyan

India's flagship national water conservation campaign, launched in 2019.

The Mandate	A time-bound, mission-mode campaign focused on water-stressed districts, run jointly by central and state governments.
The Five Focus Areas	Water conservation and rainwater harvesting; renovation of traditional water bodies; reuse and recharge structures; watershed development; and intensive afforestation.
Why It Matters Here	It operationalizes the exact interventions in this section — rainwater harvesting, recharge, and precision water management — at national scale.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Roughly what share of India's total water use goes to agriculture, making precision irrigation such a high-leverage intervention?

A About 25%

B About 50%

C About 85% ✓

D About 10%

ANSWER & WHY

C. About 85%

Agriculture accounts for roughly 85% of India's total water consumption, primarily through irrigation — making even modest efficiency gains from drip and precision irrigation enormously significant at national scale.

SECTION 3 OF 4

Waste Reduction

The hierarchy of what to do with material — long before it becomes something to throw away.

The Waste Hierarchy

Not all waste management options are equal — this hierarchy ranks them from most to least preferred.

1. Refuse & Reduce — Avoid creating the waste in the first place — the only tier that eliminates impact entirely.

2. Reuse — Use a product again, in its original form, for the same or a different purpose.

3. Recycle — Break a material down and remanufacture it into something new.

4. Recover — Extract energy from waste that cannot be reused or recycled (e.g., waste-to-energy).

5. Dispose — Landfill — the least preferred option, used only when nothing else is possible.

SECTION 3 · WORKED EXAMPLE

Calculating a City's Landfill Diversion Rate

GIVEN

- A city generates 1,000 tonnes of municipal solid waste per day
- Of this: 350 tonnes is composted, 150 tonnes is recycled
- The remaining 500 tonnes goes to landfill

FORMULA

Diversion Rate (%) = (Waste Diverted from Landfill ÷ Total Waste Generated) × 100

STEP-BY-STEP CALCULATION

1. Total waste diverted: composted + recycled = 350 + 150 = 500 tonnes
2. Diversion rate: (500 ÷ 1,000) × 100 = 50%

50% Diversion

Exactly half of this city's waste is being kept out of the landfill — the standard metric cities and campuses use to track waste management progress.

Extended Producer Responsibility (EPR)

A policy principle that shifts the cost and responsibility for waste back onto the companies that created it.

The Core Idea	Producers are made financially and physically responsible for the end-of-life management of the products and packaging they place on the market.
India's E-Waste (Management) Rules, 2022	Mandate producers of electronics to meet specific e-waste collection and recycling targets, obtained through registered recyclers.
India's Plastic Waste Management Rules	Require producers, importers, and brand owners of plastic packaging to meet EPR targets for collection and recycling.
Why It Changes Behaviour	When producers bear disposal costs, they have a direct financial incentive to design products that are easier and cheaper to recycle.

◆ INTERACTIVE

CLASSIFY THE ACTION

Where Does This Sit in the Waste Hierarchy?

For each real action, decide which tier of the waste hierarchy it belongs to.

1. Refilling a glass bottle instead of buying a new one
2. Melting down aluminium cans to make new cans
3. Sending unsorted mixed waste directly to a landfill
4. Designing packaging to use less material from the start
5. Burning non-recyclable waste to generate electricity
6. Donating used furniture instead of throwing it away

REDUCE/REUSE	RECYCLE/RECOVER	DISPOSE
1	2	3
4	5	
6		

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Under Extended Producer Responsibility (EPR), who is made financially responsible for a product's end-of-life management?

A The consumer who purchased the product

B The local municipal government

C The producer or brand owner who placed the product on the market ✓

D Recycling companies exclusively

ANSWER & WHY

C. The producer or brand owner who placed the product on the market

EPR shifts responsibility for end-of-life management — collection, recycling, safe disposal — back onto the producer or brand owner, creating a direct financial incentive to design products that are easier and cheaper to recycle.

SECTION 4 OF 4

The Circular Economy

The systems-level model that ties energy, water, and waste together — permanently, not as separate projects.

Three Principles of a Circular Economy

As defined by the Ellen MacArthur Foundation — the most widely cited framework in the field.

Design Out Waste & Pollution

Waste is treated as a design flaw, not an inevitability — products and systems are designed from the outset to prevent it.

Keep Products & Materials in Use

Through reuse, repair, remanufacturing, and recycling — extending value for as long as physically possible.

Regenerate Natural Systems

Circular systems actively rebuild natural capital — returning nutrients to soil, restoring ecosystems — not just minimizing harm.

The Technical Cycle vs. the Biological Cycle

The Ellen MacArthur Foundation's "butterfly diagram" splits circularity into two distinct loops, based on what kind of material is involved.

The Technical Cycle

- Applies to durable, manufactured materials: metals, plastics, glass, electronics.
- Circulates through maintenance, reuse, refurbishment, remanufacturing, and recycling.
- Goal: keep the material in a manufactured loop indefinitely, never returning to raw extraction.

The Biological Cycle

- Applies to renewable, biodegradable materials: food waste, natural fibers, wood.
- Circulates through composting and anaerobic digestion, safely returning nutrients to the soil.
- Goal: regenerate natural systems, not just avoid landfill.

SECTION 4 · WHERE THE WORLD ACTUALLY STANDS

The Circularity Gap

6.9%

The share of materials in the global economy that are cycled back — down from 9.1% in 2018

93.1%

Share of all materials entering the global economy that still come from virgin extraction

€25.4 Tn

The "Value Gap": economic value lost annually to linear resource use — nearly 31% of global GDP

Despite two decades of circular economy advocacy, the world has gotten measurably less circular, not more — the gap is the opportunity.

Source: Circle Economy & Deloitte, Circularity Gap Report 2025 & 2026.

SECTION 4 · WHERE THE VALUE ACTUALLY LEAKS

The Five Pathways of Value Loss

The Circularity Gap Report 2026 breaks the €25.4 trillion Value Gap into five specific leaks in the system.

End-of-Life Waste (€10.0 Tn)

The single largest leak: products landfilled or incinerated instead of refurbished, reused, or recycled.

Energy Losses (€8.7 Tn)

Energy lost during raw material extraction, inefficient grid transmission, and poor thermal regulation in buildings.

Food Losses & Waste

Edible food lost across production, distribution, and consumption before it ever reaches — or after it leaves — a plate.

Processing Losses

Material lost as offcuts, scrap, and by-products during manufacturing that are never captured or reused.

Premature Deterioration

Buildings, infrastructure, and equipment retired or replaced well before the end of their useful physical life.

◆ INTERACTIVE

CLASSIFY THE MATERIAL

Technical Cycle or Biological Cycle?

For each material, decide which of the two circular economy loops it belongs to.

1. Aluminium cans
2. Food scraps and vegetable peels
3. Smartphone components
4. Garden and yard waste
5. Glass bottles
6. Untreated wood offcuts

TECHNICAL CYCLE	BIOLOGICAL CYCLE
1	2
3	4
5	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

According to the Circularity Gap Report, what is the single largest of the five pathways of value loss in the global economy?

A Food losses and waste

B Processing losses

C End-of-life waste ✓

D Premature deterioration of buildings

ANSWER & WHY

C. End-of-life waste

End-of-life waste — products landfilled or incinerated instead of refurbished, reused, or recycled — is the largest single pathway of value loss, estimated at roughly €10.0 trillion annually in the Circularity Gap Report 2026.

BRINGING IT TOGETHER

One Systems Problem, Not Four Separate Ones

Energy, water, and waste are not independent projects — the circular economy is the framework that shows why.

Energy Efficiency	Reduces the energy embedded in every material extracted, processed, and moved — the "energy losses" pathway in the Value Gap.
Water Conservation	Recycled water and precision irrigation are themselves circular loops — keeping a resource in use instead of extracting more.
Waste Reduction	The waste hierarchy is a miniature version of the circular economy — refuse and reuse before recycle, recycle before dispose.
The Circular Economy	The framework that shows these three are not separate initiatives — they are one system, viewed from three different points.

◆ INTERACTIVE

APPLY IT ALL

"Pick a building or campus you know well. Where would you find the fastest-payback energy fix, the biggest water-saving opportunity, and the worst waste leak — all three?"

This is exactly how a resource management audit actually starts.

⏱ 4 MIN

OPERATIONAL SUSTAINABILITY

Sustainable Supply Chains

Integrating environmental considerations into sourcing, production, and distribution

You already know 70–90% of a company's footprint lives in its supply chain.
Today is about what to actually do about it.

Three stages — sourcing, production, distribution — and the specific interventions, standards, and case studies that make each one measurably more sustainable.

Following the Product, Upstream to Downstream

Sustainable Supply Chain Management (SSCM): integrating environmental and social criteria into every decision along a product's journey — not just at the factory gate.



01

Sourcing

Where raw materials come from, and under what conditions.



02

Production

How materials become products — and at what environmental cost.



03

Distribution

How products move to market — often the most visible emissions of all.



04

Measuring the Chain

Mapping, tracking, and reporting the whole system end to end.

SECTION 1 OF 4

Sustainable Sourcing

Where raw materials come from, and under what conditions — the origin point of most supply chain risk.

SECTION 1 · DEFINITION

What Is Sustainable Sourcing?

The furthest-upstream link in the chain — and, per the GHG Protocol, usually the largest share of a company's real footprint.

The Working Definition

Procuring raw materials and components in ways that account for environmental impact, labor conditions, and long-term resource availability — not price alone.

Why Upstream Matters Most

Categories 1 and 4 of Scope 3 (purchased goods, upstream transportation) are frequently a company's single largest emissions category — often larger than its own operations.

The Core Challenge

A company rarely owns its suppliers — sustainable sourcing means influencing behavior it cannot directly control, several tiers removed from its own operations.

SCOPE 3: Categories:

Everything that happens before a product or service reaches your organization.



1. Purchased Goods & Services

Emissions from producing everything your organization buys or acquires.



2. Capital Goods

Emissions from producing purchased long-term assets — machinery, buildings, equipment.



3. Fuel- & Energy-Related Activities

Upstream emissions of fuel and energy not already counted in Scope 1 or 2.



4. Upstream Transportation & Distribution

Third-party transport of purchased products to your organization.



5. Waste Generated in Operations

Emissions from third-party disposal and treatment of your operational waste.



6. Business Travel

Employee travel for business in vehicles not owned by the organization — flights, trains, taxis.



7. Employee Commuting

Emissions from employees' travel between home and work.



8. Upstream Leased Assets

Emissions from operating assets you lease, not already in Scope 1 or 2.

Four Categories of Sourcing Risk

The specific failure modes that make upstream sourcing the riskiest link in most supply chains.



Deforestation-Linked Commodities

Palm oil, soy, cattle, and timber are consistently linked to tropical deforestation when sourced without traceability.



Conflict Minerals

Tin, tungsten, tantalum, and gold sourced from conflict-affected regions can finance armed groups and human rights abuses.



Forced & Child Labour

Complex, multi-tier supply chains can obscure labor abuses far removed from the buying company's direct view.



Biodiversity & Habitat Loss

Extraction of raw materials — mining, fishing, logging — can degrade ecosystems far from where the final product is sold.

Certifications That Address Sourcing Risk

Third-party certification is the primary tool buyers use to verify claims they cannot check themselves.

Fairtrade

Certifies fair prices and labor standards for commodities like coffee, cocoa, and cotton, verified by independent audit.

Forest Stewardship Council (FSC)

Certifies wood and paper products from responsibly managed forests, with full chain-of-custody tracking.

Rainforest Alliance

Certifies farms meeting environmental, social, and economic sustainability criteria, common in coffee, tea, and cocoa.

Responsible Minerals Initiative

Audits smelters and refiners to verify minerals are not sourced from conflict-affected or high-risk areas.

◆ INTERACTIVE

CASE STUDY

When Sourcing Oversight Fails: Rana Plaza



The Rana Plaza Collapse, Dhaka, Bangladesh (April 24, 2013)

An eight-story building housing five garment factories collapsed, killing 1,134 workers and injuring approximately 2,500 more — the deadliest disaster in the history of the garment industry.

The Sourcing Failure

The factories supplied major global apparel brands, whose supplier audits had never identified the building's fundamental structural defects.

The Warning Ignored

Visible cracks were discovered the day before the collapse; workers were ordered back to their 13–14 hour shifts regardless.

The Systemic Response

The legally binding Accord on Fire and Building Safety in Bangladesh was signed within a month by over 200 global brands and retailers.

The Lesson

Supplier audits that check labor and quality standards, but not structural safety, can miss the most catastrophic risks entirely.

◆ INTERACTIVE

THINK & SHARE

"A brand's supplier audit checks fire exits and pay stubs, but not structural engineering. What other kinds of 'invisible' risk might a standard audit checklist miss today?"

Think beyond garments — electronics, food, minerals.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

For most companies, which GHG Protocol Scope 3 categories are most directly addressed by sustainable sourcing practices?

A Category 11 (Use of sold products) and Category 12 (End-of-life)

B Category 1 (Purchased goods & services) and Category 4 (Upstream transportation) ✓

C Category 6 (Business travel) only

D Sourcing has no connection to Scope 3 at all

ANSWER & WHY

B. Category 1 (Purchased goods & services) and Category 4 (Upstream transportation)

Sustainable sourcing addresses the upstream categories of Scope 3 most directly — Category 1 (purchased goods & services) and Category 4 (upstream transportation) — which are frequently a company's single largest emissions categories.

SECTION 2 OF 4

Sustainable Production

How raw materials become products — and the environmental cost embedded in every design choice.

SECTION 2 · DEFINITION

What Is Sustainable Production?

Manufacturing that minimizes environmental impact per unit of output — not simply "cleaner factories."

The Working Definition

Producing goods using processes and materials that minimize energy use, water use, waste, and toxic emissions relative to the value created.

Cleaner Production, Not End-of-Pipe

The modern approach prevents pollution and waste at the source — in process and material design — rather than treating waste after it's already created.

Why It's a Design Problem

The vast majority of a product's environmental impact is locked in at the design stage, long before a single unit is manufactured.

SECTION 2 · THE CORE TOOL

Life Cycle Assessment (LCA)

The standardized method for actually measuring a product's environmental impact — from raw material to disposal.

Cradle-to-Gate	Measures impact from raw material extraction through to the factory gate — the boundary most manufacturers can measure directly.
Cradle-to-Grave	Extends the assessment through use and disposal — the full picture of a product's lifetime impact.
Cradle-to-Cradle	Extends further still: designed explicitly so the product's materials become the raw material for the next product cycle.
Why It Matters Here	LCA is what lets a company compare two production or material choices quantitatively, instead of guessing which is "greener."

SECTION 2 · WORKED EXAMPLE

Comparing Packaging Materials with LCA Thinking

GIVEN

- Two packaging options, each weighing 50 g (0.050 kg) per unit
- Option A — Virgin plastic: embodied emissions of 3.5 $kgCO_2e$ per kg of material produced
- Option B — Recycled plastic (rPET): embodied emissions of 1.2 $kgCO_2e$ per kg of material produced

FORMULA

Embodied Emissions per Unit = Weight (kg) × Emission Factor ($kgCO_2e$/kg)

STEP-BY-STEP CALCULATION

1. Option A: 0.050 kg × 3.5 $kgCO_2e$/kg = 0.175 $kgCO_2e$ per unit
2. Option B: 0.050 kg × 1.2 $kgCO_2e$/kg = 0.060 $kgCO_2e$ per unit
3. Savings per unit: 0.175 − 0.060 = 0.115 $kgCO_2e$
4. At 1 million units produced per year: 0.115 kg × 1,000,000 = 115,000 kg = 115 tonnes CO_2e saved

115 t CO_2e / Year

A single material substitution — virgin to recycled plastic — saves the equivalent of many households' annual emissions, without changing the product itself.

Four Cleaner Production Techniques

Concrete interventions that reduce environmental impact at the point of manufacture.



Energy & Water Efficiency

Optimizing industrial processes to use less energy and water per unit of output — often the fastest payback in a factory.



Waste Minimization

Redesigning processes to generate less scrap, off-cuts, and by-product in the first place — not just recycling more of it.



Safer Chemical Substitution

Replacing hazardous solvents, dyes, and inputs with less toxic alternatives that achieve the same manufacturing result.



Process Optimization

Fine-tuning temperatures, cycle times, and equipment settings to cut energy and material use without changing the product.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

What is the key difference between "cradle-to-gate" and "cradle-to-cradle" in Life Cycle Assessment?

A They measure exactly the same boundary

B Cradle-to-gate stops at the factory; cradle-to-cradle is designed so end-of-life materials become new raw material ✓

C Cradle-to-cradle only applies to food products

D Cradle-to-gate includes disposal, cradle-to-cradle does not

ANSWER & WHY

B. Cradle-to-gate stops at the factory; cradle-to-cradle is designed so end-of-life materials become new raw material

Cradle-to-gate measures impact only up to the factory gate. Cradle-to-cradle is the most ambitious boundary — the product and its materials are explicitly designed to become the raw material for a future product cycle, embodying circular economy principles directly in the LCA scope.

SECTION 3 OF 4

Sustainable Distribution

How products move to market — often the most visible, and most easily reducible, link in the chain.

SECTION 3 · THE SCALE OF THE PROBLEM

Distribution's Environmental Footprint

~1/5

Share of global CO_2 emissions attributable to the transport sector as a whole

4–6

Category 4 (upstream) and Category 9 (downstream) transportation — two of the most under-measured Scope 3 categories

75×

How much more emissions-intensive air freight is than sea freight, for the same cargo and distance

Distribution choices are unusual in sustainability: the lowest-carbon option is frequently also the lowest-cost option — a rare true win-win.

Source: Illustrative figures based on published freight-transport emission factors (e.g., DEFRA, ICCT).

SECTION 3 · WORKED EXAMPLE

Comparing Freight Transport Modes

GIVEN

- Shipment: 20 tonnes of goods, travelling 6,000 km
- Sea freight emission factor: ~10 gCO_2e per tonne-km
- Air freight emission factor: ~600 gCO_2e per tonne-km

FORMULA

Emissions (kg) = Weight (t) × Distance (km) × Emission Factor (g/t·km) ÷ 1,000

STEP-BY-STEP CALCULATION

1. Sea freight: 20 × 6,000 × 10 ÷ 1,000 = 1,200 kg ≈ 1.2 tonnes CO_2e
2. Air freight: 20 × 6,000 × 600 ÷ 1,000 = 72,000 kg = 72 tonnes CO_2e
3. Ratio: 72 ÷ 1.2 = 60× more emissions for the identical shipment, moved by air instead of sea

60× Difference

The mode choice alone — sea versus air, for the exact same cargo — changes the shipment's footprint by a factor of 60. Mode shift is one of the highest-leverage decisions in distribution.

Five Levers of Sustainable Distribution

Interventions that reduce distribution's footprint without necessarily changing what is sold.

Modal Shift

Moving freight from air or road to sea or rail wherever transit time allows — the single highest-leverage lever available.

Route Optimization

Software-driven routing that reduces total distance travelled and empty return trips ("deadheading").

Packaging Reduction

Lighter, right-sized packaging reduces both material use and the weight (and therefore fuel) needed to ship it.

Warehouse Efficiency

Locating distribution centers closer to demand, and using energy-efficient warehouse operations, cuts both distance and energy use.

Reverse Logistics

Efficient systems for returns, repairs, and end-of-life collection keep products and materials circulating rather than being discarded.

◆ INTERACTIVE

CLASSIFY THE INTERVENTION

Sourcing, Production, or Distribution?

For each real intervention, decide which stage of the supply chain it primarily belongs to.

1. Requiring FSC certification for all wood suppliers
2. Switching a factory from virgin to recycled plastic input
3. Shifting freight from air to sea for non-urgent shipments
4. Auditing a mineral supplier against conflict-minerals standards
5. Redesigning a manufacturing process to cut water use
6. Right-sizing packaging to reduce shipping weight

SOURCING	PRODUCTION	DISTRIBUTION
1	2	3
4	5	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the worked example, why did the air freight shipment produce roughly 60 times more emissions than the identical sea freight shipment?

A The air shipment travelled a much longer distance

B The air shipment carried more weight

C Air freight has a much higher emission factor (gCO_2e per tonne-km) than sea freight ✓

D There is no real difference between air and sea freight emissions

ANSWER & WHY

C. Air freight has a much higher emission factor (gCO_2e per tonne-km) than sea freight

Both shipments had identical weight and distance — the entire 60× difference comes from the emission factor itself: air freight emits far more CO_2e per tonne-kilometre than sea freight, due to the energy intensity of air transport.

SECTION 4 OF 4

Measuring & Managing the Chain

You cannot manage what you cannot see — the visibility problem underneath every claim in this lecture.

Supply Chain Tiers: Why Visibility Collapses Fast

Most companies can see their direct suppliers clearly — and almost nothing beyond that.

Tier 1

Direct suppliers the company contracts with and pays directly.

Usually well audited and visible.

Tier 2

Suppliers to your Tier 1 suppliers — e.g., the fabric mill supplying a garment factory.

Visibility drops sharply here.

Tier 3+

Raw material extraction — mines, farms, forests — often multiple contractual layers removed.

Frequently invisible to the buying company entirely.

SECTION 4 · CONNECTING BACK

The Whole Lecture Is a Scope 3 Map

Every stage covered today maps directly onto specific GHG Protocol Scope 3 categories.

Sourcing	Cat. 1 — Purchased goods & services · Cat. 2 — Capital goods
Production	Cat. 3 — Fuel- & energy-related activities · Cat. 5 — Waste generated in operations
Distribution	Cat. 4 — Upstream transportation · Cat. 9 — Downstream transportation
End-of-Life (Beyond Today)	Cat. 11 — Use of sold products · Cat. 12 — End-of-life treatment

◆ INTERACTIVE

CASE STUDY

When Supplier Engagement Works at Scale: Project Gigaton



Walmart's Project Gigaton (Launched 2017)

Walmart invited its more than 100,000 suppliers to voluntarily reduce or avoid 1 billion metric tons (a gigaton) of greenhouse gas emissions from its supply chain by 2030 — targeting energy, waste, packaging, transportation, nature, and product design.

The Result

In February 2024, Walmart announced suppliers had collectively reported over 1 billion metric tons reduced or avoided — six years ahead of the 2030 target.

The Scale

More than 5,000 suppliers ultimately participated — described as one of the largest private-sector supply chain sustainability consortia ever assembled.

The Method

Not top-down mandates — a shared toolkit, education, financing support, and public recognition for supplier progress.

The Honest Caveat

The same year, Walmart disclosed its own direct (Scope 1 & 2) operational emissions rose 3.9% — supply chain success and operational progress are not the same thing.

Technology for Supply Chain Traceability

The engineering opportunity underneath everything in this lecture — visibility itself is a data problem.

Blockchain-Based Traceability	Distributed ledgers can record a product's custody chain immutably, from raw material to shelf, making origin claims independently verifiable.
Satellite & Remote Sensing	Monitoring platforms can detect deforestation or land-use change linked to specific supply chain regions in near real time.
Supplier Data Platforms	Software that aggregates supplier-reported emissions, labor, and safety data into a single auditable system, replacing scattered spreadsheets.
Why This Is Your Opportunity	Every technology on this slide is still immature relative to the scale of the problem — this is one of the most under-built areas in all of climate tech.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Why does supply chain visibility typically collapse beyond Tier 1 suppliers?

A Tier 2 and Tier 3 suppliers are always located in different countries

B Companies have direct contracts only with Tier 1 suppliers, and visibility depends on those suppliers disclosing their own suppliers in turn ✓

C Tier 2 and Tier 3 suppliers are legally required to remain anonymous

D Visibility does not actually decrease at any tier

ANSWER & WHY

B. Companies have direct contracts only with Tier 1 suppliers, and visibility depends on those suppliers disclosing their own suppliers in turn

A company's direct contractual relationship is with its Tier 1 suppliers. Seeing further upstream requires those suppliers to voluntarily disclose their own suppliers — a chain of disclosure that gets weaker, and less enforceable, at every additional tier.

BRINGING IT TOGETHER

One Chain, Three Interventions, One Measurement System

Sourcing, production, and distribution are not separate sustainability projects — they are three points on the same value chain, measured by the same framework.

Sustainable Sourcing	Reduces risk and impact before a material ever enters the factory — certifications and audits are the tools.
Sustainable Production	Reduces impact per unit made — Life Cycle Assessment is the tool that makes design choices comparable.
Sustainable Distribution	Reduces impact per unit moved — mode shift is the single highest-leverage lever available.
Measurement & Visibility	Ties all three together into one Scope 3 inventory — and is the actual bottleneck most companies face today.

◆ INTERACTIVE

APPLY IT ALL

"Pick a product you own. Trace it backward: where was it sourced, produced, and distributed from — and which of today's three interventions would improve it most?"

This is exactly how a real supply chain sustainability audit begins.

⏱ 4 MIN

Life Cycle Assessment (LCA)

The tools and techniques that let us measure a product's environmental impact, cradle to grave

"Is a plastic bag worse than a paper bag?" LCA is the discipline that turns that argument into an actual number.

Every "greener" claim you've ever heard about a product rests on a Life Cycle Assessment somewhere — today you learn how those numbers actually get made.

BY THE END OF THIS LECTURE

You will be able to...



Define

State the ISO 14040/14044 four-phase LCA framework and the functional unit concept precisely.



Calculate

Build a simple life cycle inventory and compute a characterized impact score by hand.



Compare

Recognize the major impact categories and the software tools practitioners actually use.



Critique

Identify where LCA results can mislead, and why the same product can get two different answers.

TODAY'S JOURNEY

The Four Phases of an ISO-Compliant LCA

Life Cycle Assessment traces back to a 1969 Coca-Cola packaging study — standardized globally since ISO 14040 was first published in 1997.



01

Goal & Scope

Define why you're measuring, and exactly what "one unit" of comparison means.



02

Inventory (LCI)

Quantify every material and energy input, and every emission and waste output.



03

Impact Assessment (LCIA)

Translate raw inventory data into actual environmental impact scores.



04

Interpretation

Identify what the numbers actually mean — and where they can be trusted.

SECTION 1 OF 4

Goal & Scope Definition

Before any measurement begins: what exactly are we comparing, and against what unit?

SECTION 1 · WHERE IT ALL STARTED

A Short History of LCA

The methodology is older than the environmental movement that now relies on it.

1969 — The Coca-Cola Study

Coca-Cola commissioned the Midwest Research Institute to compare glass, plastic, and metal beverage containers — a "Resource and Environmental Profile Analysis" (REPA), the acknowledged precursor to modern LCA.

1990 — A Formal Definition

The Society of Environmental Toxicology and Chemistry (SETAC) formally defined LCA as an objective process for evaluating environmental burdens.

1997–2006 — ISO Standardization

The ISO 14040 series was first published in 1997 and finalized in its current form (ISO 14040:2006 and ISO 14044:2006), the global standard used today.

SECTION 1 · THE FIRST PHASE

What Is Goal & Scope Definition?

The most consequential phase of an LCA — decisions made here shape every number that follows.

Setting the Goal

Why is this study being done, and who is the intended audience? A study for internal design decisions differs from one meant for public marketing claims.

Setting the Scope

What system boundaries, level of detail, and assumptions will the study use? This is where the study's credibility is either built or undermined.

Why It Comes First

Every later phase depends entirely on choices made here — get the goal and scope wrong, and no amount of careful data collection afterward can fix it.

SECTION 1 · THE KEY CONCEPT

The Functional Unit

The single most important, and most frequently misunderstood, concept in all of LCA.

The Definition

A quantified description of the performance a product system delivers — the reference unit that makes fair comparison possible.

Why "One Product" Isn't Enough

"One hand dryer" and "one paper towel" cannot be compared directly — but "drying one pair of hands, once" can. The functional unit must describe the function, not the object.

A Classic Failure Mode

Comparing "one reusable cotton bag" to "one plastic bag" ignores that the cotton bag must be reused dozens or hundreds of times to deliver the same function as many single-use bags — the functional unit exposes this immediately.

SECTION 1 · WORKED EXAMPLE

Comparing Products via a Shared Functional Unit

GIVEN

- Functional unit: "Drying one pair of hands, once"
- Option A — Electric dryer: 1.8 kW rating, used for 25 seconds per event
- Option B — Paper towel: 2 sheets per event, 10 g CO_2e embodied per sheet

FORMULA

Emissions per Functional Unit = Energy or Material Used per Event × Emission Factor
(Grid factor used: 0.7 $kgCO_2e$/kWh)

STEP-BY-STEP CALCULATION

1. Dryer energy per event: 1.8 kW × (25 ÷ 3,600) hr = 0.0125 kWh
2. Dryer emissions: 0.0125 kWh × 0.7 $kgCO_2e$/kWh = 0.00875 kg = 8.75 g CO_2e
3. Paper towel emissions: 2 sheets × 10 g CO_2e/sheet = 20 g CO_2e

8.75 g vs. 20 g

In this simplified model, the electric dryer's footprint per use is roughly 2.3× lower than paper towels — a comparison only possible because both were reduced to the same functional unit.

SECTION 1 · SETTING THE EDGES

Four System Boundary Choices

How far along the life cycle the assessment actually looks — a decision made once, in scope definition, that shapes everything after.



Gate-to-Gate

Covers only a single process step inside one facility — the narrowest possible boundary.



Cradle-to-Gate

From raw material extraction through to the factory gate, stopping before distribution and use.



Cradle-to-Grave

The full journey: raw material, production, distribution, use, and final disposal.



Cradle-to-Cradle

The most ambitious boundary: end-of-life materials are explicitly designed to become the next product's raw material.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Why can't "one reusable cotton tote bag" be fairly compared to "one single-use plastic bag" without a functional unit?

A Cotton bags are always more expensive

B A cotton bag must be reused many times to deliver the same total carrying function as many single-use bags ✓

C Plastic bags cannot be measured with LCA at all

D There is no actual difference between the two products

ANSWER & WHY

B. A cotton bag must be reused many times to deliver the same total carrying function as many single-use bags

The functional unit forces a fair comparison: since a cotton bag must be reused many times to match the total "carrying" function of the plastic bags it replaces, the honest comparison is "X reusable bag uses" vs. "X single-use bags" — not one unit of each.

SECTION 2 OF 4

Life Cycle Inventory (LCI)

Turning a defined scope into an actual list of every input and output in the system.

SECTION 2 · THE SECOND PHASE

What Is the Life Cycle Inventory?

The most data-intensive phase of any LCA: quantifying literally everything that crosses the system boundary.



INPUTS

- Raw materials extracted or purchased
- Energy consumed (electricity, fuel)
- Water withdrawn from the environment



OUTPUTS

- Products and co-products created
- Emissions to air, water, and soil
- Solid waste and by-products generated

Primary vs. Secondary Data

LCA practitioners rarely have direct measurements for every single process — they combine both types of data.

Primary Data

Directly measured or metered data from the actual facility or supply chain being studied — the most accurate, but the most expensive and time-consuming to collect.

Secondary Data

Data from published databases, industry averages, or literature, used to fill gaps where primary data isn't available or practical.

Databases in Practice

Ecoinvent is the most widely used commercial LCI database worldwide, providing thousands of standardized process datasets across industries.

SECTION 2 · A GENUINE METHODOLOGICAL PROBLEM

The Allocation Problem

What happens when one process produces more than one useful output?

Example: an oil refinery produces petrol, diesel, and jet fuel from the same barrel of crude oil in one process. How much of the refinery's total emissions belongs to each fuel?

Mass Allocation

Split impacts in proportion to the physical mass of each co-product — simple, but ignores that products have very different economic and functional value.

Economic Allocation

Split impacts in proportion to each co-product's market value — reflects why the process exists, but shifts with market prices.

System Expansion

Avoid allocation entirely by expanding the system boundary to credit the process for displacing an alternative product — the ISO-preferred approach, but often harder to apply.

Allocation Problem

Many industrial processes generate multiple useful products.

Suppose a process consumes: 1000 MJ

and produces:

- Product A = 600 kg
- Product B = 400 kg

How should the environmental burden be divided?

This is known as **allocation**.

Mass allocation: $\text{Allocation}_A = 600/(600+400)=0.60$

Therefore: $\text{Impact}_A=0.60I_{total}$

Economic allocation

If Product A has substantially greater economic value, allocation can instead be based on revenue.

$f_A= m_A*P_A /(m_A*P_A+m_B*P_B$

where P represents price.

Different allocation methods can produce significantly different results. Therefore allocation assumptions must be clearly reported and, where appropriate, tested through sensitivity analysis.

SECTION 2 · WORKED EXAMPLE

Building a Life Cycle Inventory Entry

GIVEN

- 1,180 kg of raw plastic resin is fed into a process to manufacture 1,000 kg of finished product
- The process consumes 450 kWh of electricity
- Grid emission factor: 0.7 $kgCO_2e/kWh$

FORMULA

Process Scrap = Material Input − Product Output
Process Emissions = Energy Used × Emission Factor

STEP-BY-STEP CALCULATION

1. Scrap generated: 1,180 kg − 1,000 kg = 180 kg of solid waste per 1,000 kg of product
2. Process emissions: 450 kWh × 0.7 $kgCO_2e/kWh$ = 315 $kgCO_2e$ per 1,000 kg of product
3. Per functional unit (1 kg of product): 0.18 kg scrap and 0.315 $kgCO_2e$ — both become entries in the inventory table

0.18 kg + 0.315 kg CO_2e

Per kilogram of product: 0.18 kg of solid waste and 0.315 kg of CO_2e emissions — two concrete inventory entries built directly from a simple mass and energy balance.

◆ INTERACTIVE

THINK & SHARE

"A dairy produces both milk and leather (from the same cattle). Which allocation method — mass, economic, or system expansion — would you choose to split the farm's emissions between them, and why?"

There is no universally correct answer — that is exactly why allocation is contested.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Which allocation approach does the ISO 14044 standard prefer, when it can practically be applied?

A Mass allocation

B Economic allocation

C System expansion (avoiding allocation by crediting displaced products) ✓

D ISO expresses no preference at all

ANSWER & WHY

C. System expansion (avoiding allocation by crediting displaced products)

ISO 14044 prefers avoiding allocation altogether through system expansion wherever practical — crediting a process for the alternative product it displaces — turning to mass or economic allocation only when system expansion isn't feasible.

SECTION 3 OF 4

Life Cycle Impact Assessment (LCIA)

Turning a long list of emissions and resource use into actual, comparable impact scores.

SECTION 3 · THE THIRD PHASE

What Is Life Cycle Impact Assessment?

The step that turns "2 kg of CO_2 and 0.5 g of SO_2" into something a decision-maker can actually act on.

Classification

Sorting each inventory flow into the impact category (or categories) it contributes to — e.g., CO_2 goes into "climate change."

Characterization

Converting each flow into a common unit within its category using a characterization factor — exactly how GWP converts CH_4 and N_2O into CO_2e.

Normalization & Weighting (Optional)

Comparing category scores against a reference (e.g., per-capita global impact), and optionally combining categories into a single overall score — the most subjective, most debated steps.

SECTION 3 · THE SCORECARD

Six Major Impact Categories

GWP is the most famous impact category — but a rigorous LCA reports several others in parallel.

Global Warming Potential

Climate change impact, expressed in kg CO_2-equivalent — the category covered in depth in our carbon footprint lecture.

Ozone Depletion Potential

Damage to the stratospheric ozone layer, expressed in kg CFC-11-equivalent.

Acidification Potential

SO_2 and NOx emissions that cause acid rain and ecosystem damage, expressed in kg SO_2-equivalent.

Eutrophication Potential

Excess nitrogen and phosphorus causing algal blooms and oxygen depletion in water bodies.

Human Toxicity Potential

Potential health effects from chemical exposure across the life cycle.

Water Scarcity Footprint

Freshwater consumption weighted by the local water stress of where it was withdrawn.

SECTION 3 · WORKED EXAMPLE

Characterization: From Raw Emissions to One Impact Score

GIVEN

- A product's inventory (per functional unit) includes: 2 kg CO_2, 0.01 kg CH_4, and 0.002 kg N_2O
- GWP-100 characterization factors (IPCC AR6): CO_2 = 1, CH_4 = 27, N_2O = 273

FORMULA

Impact Score ($kgCO_2e$) = Σ [Mass of Substance × Characterization Factor]

STEP-BY-STEP CALCULATION

1. CO_2: 2 kg × 1 = 2 $kgCO_2e$
2. CH_4: 0.01 kg × 27 = 0.27 $kgCO_2e$
3. N_2O: 0.002 kg × 273 = 0.546 $kgCO_2e$
4. Total GWP impact score: 2 + 0.27 + 0.546 = 2.816 $kgCO_2e$

2.816 kg CO_2e

This is literally what "characterization" means in LCIA — the exact same GWP-weighting logic from our carbon footprint calculation lecture, now applied inside a formal LCA.

◆ INTERACTIVE

MATCH THE IMPACT

Which Impact Category Does This Belong To?

For each emission or activity, identify the impact category it primarily contributes to.

1	CFC refrigerant leaking from old equipment
2	Nitrogen fertilizer runoff into a river
3	SO_2 emissions from a coal power plant
4	CO_2 released from fuel combustion
5	Heavy metal discharge into soil
6	Excessive groundwater extraction in a dry region

CLIMATE / OZONE	ACID / NUTRIENT	TOXICITY / WATER
1	2	5
4	3	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In LCIA, what does the "characterization" step actually do?

A. It collects the raw inventory data from suppliers

B. It converts each inventory flow into a common unit within its impact category using a characterization factor ✓

C. It decides which company should conduct the study

D. It replaces the need for a functional unit

ANSWER & WHY

B. It converts each inventory flow into a common unit within its impact category using a characterization factor

Characterization applies a scientifically derived factor (like GWP for climate change) to convert each raw inventory flow into a common unit within its impact category — the step that turns a list of different substances into one comparable number per category.

SECTION 4 OF 4

Interpretation, Tools & Applications

Turning impact scores into an actual decision — and being honest about what the numbers can and cannot tell you.

SECTION 4 · THE FOURTH PHASE

What Is the Interpretation Phase?

Where the numbers finally become a conclusion — and where a good LCA is distinguished from a misleading one.

Identifying Significant Issues

Which life cycle stages, processes, or substances actually drive the results? This is often called "hotspot analysis."

Sensitivity & Uncertainty Analysis

How much do the conclusions change if key assumptions — energy mix, transport distance, data source — are varied within a plausible range?

Conclusions & Limitations

ISO 14044 explicitly requires that every LCA report state its limitations — the boundaries, data gaps, and assumptions a reader needs to judge how far to trust it.

SECTION 4 · THE TOOLKIT

LCA Software & Databases

Practitioners rarely calculate an LCA by hand at industrial scale — these are the tools that do it.



SimaPro

One of the longest-established commercial LCA software platforms, widely used in industry and consulting.



GaBi

A major commercial LCA platform with an extensive linked database, common in automotive and industrial applications.



openLCA

A free, open-source LCA software platform, lowering the barrier to entry for students, startups, and researchers.



Ecoinvent

The most widely used LCI database worldwide, providing thousands of standardized background datasets across industries and regions.

SECTION 4 · WHERE THIS ACTUALLY GETS USED

Real-World Applications of LCA

LCA is not primarily an academic exercise — these are its main real-world uses today.



Environmental Product Declarations (EPDs)

Standardized, third-party-verified reports disclosing a product's life cycle impacts, increasingly required in construction and procurement.



Product Environmental Footprint (PEF)

The European Union's harmonized LCA-based methodology, intended to make environmental claims across the EU market comparable.



Eco-Design

Using LCA results during product design itself to compare material and process choices before manufacturing begins.



Corporate Sustainability Reporting

Feeding directly into Scope 3 emissions calculations and public sustainability disclosures.

SECTION 4 · A GENUINELY CONTESTED CASE

The Paper vs. Plastic Bag Debate

One of the most frequently cited — and frequently misunderstood — LCA results in public discourse.

The Intuitive Assumption

Plastic feels disposable and paper feels natural, so paper bags are widely assumed to be the more sustainable choice.

What Multiple LCA Studies Have Found

Producing paper bags is typically far more energy-, water-, and material-intensive than producing single-use plastic bags — meaning a paper bag can have a higher climate impact per bag manufactured.

The Reusable Bag Twist

Reusable bags (cotton, especially organic cotton) generally have the highest impact to manufacture of all — some published LCA studies estimate they must be reused many dozens to several thousand times before their total impact matches a single-use plastic bag used once.

The Real Lesson

The "obviously greener" choice is frequently wrong when it hasn't been through a rigorous functional-unit comparison — this is exactly why LCA exists as a discipline.

SECTION 4 · BEING HONEST ABOUT THE METHOD

Critiques & Limitations of LCA

A rigorous course treats LCA the same way it treated the Triple Bottom Line and shared value: useful, and imperfect.

Data Quality & Availability

Reliable, representative data is expensive and scarce, especially for supply chains in developing regions — forcing reliance on generic secondary data.

Subjectivity in Boundary-Setting

Where the system boundary is drawn, and what gets excluded via "cut-off criteria," can swing results significantly without technically breaking any rule.

The Allocation Problem, Revisited

As shown in Section 2, three defensible allocation methods can produce three different answers for the same co-product system.

"LCA Shopping"

Because so many methodological choices are left to the practitioner, it is possible to select assumptions that favor a predetermined conclusion — a real risk when a company funds its own study.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

What does ISO 14044 require every LCA report to explicitly disclose, to protect against misleading conclusions?

A The exact profit margin of the product studied

B The study's limitations, boundaries, data sources, and key assumptions ✓

C The names of every employee involved in the study

D Nothing — ISO 14044 has no disclosure requirements

ANSWER & WHY

B. The study's limitations, boundaries, data sources, and key assumptions

ISO 14044 requires transparent disclosure of a study's limitations, system boundaries, data quality, and assumptions — precisely so that a reader can judge for themselves how far to trust the conclusions, and detect potential "LCA shopping."

BRINGING IT TOGETHER

The Four Phases, End to End

Each phase depends entirely on the one before it — and ISO 14044 makes the process explicitly iterative, not strictly linear.

1. Goal & Scope	Sets the functional unit and system boundary — every later number is only comparable within this frame.
2. Inventory (LCI)	Turns the scope into a quantified list of every input and output, resolving allocation for shared processes.
3. Impact Assessment (LCIA)	Classifies and characterizes that inventory into comparable scores across categories like GWP.
4. Interpretation	Finds the hotspots, tests the assumptions, and states the limitations — turning numbers into a decision.

◆ INTERACTIVE

APPLY IT ALL

"Pick a product you own. What would its functional unit be, what allocation problem might it face, and which impact category — beyond climate — would you expect to matter most?"

This is exactly how a real LCA scoping conversation begins.

⏱ 4 MIN

The Business Case for Sustainability

Risk, opportunity, competitive advantage, stakeholder engagement, and long-term value creation

Sustainability used to be a cost center. Boards now treat it as a source of risk, capital, and competitive edge.

Today is not about ethics — it's about the actual mechanics of how sustainability moves a company's risk profile, its access to capital, and its bottom line.

From Risk to Value Creation



01

The Risk Landscape

What happens when sustainability is ignored — in real, costed terms.



02

The Opportunity Landscape

Where sustainability becomes a genuine source of competitive advantage.



03

Stakeholder Theory & Engagement

Who a company actually owes value to, and how it decides what matters.



04

Long-Term Value Creation

How all of this shows up in the price of capital itself.

SECTION 1 OF 4

The Risk Landscape

What genuinely goes wrong, in costed terms, when sustainability is treated as optional.

SECTION 1 · THE SHIFT

Why This Is No Longer Optional

Sustainability moved from a philanthropic afterthought to a core strategic and financial concern in roughly two decades.

The Old Model: Compliance & Philanthropy

For most of the 20th century, environmental and social action was treated as a cost — regulatory compliance, or corporate philanthropy separate from core strategy.

The Turning Point

Rising climate litigation, mandatory disclosure regimes (like BRSR and CSRD), extreme weather losses, and ESG-driven investment flows made sustainability a material financial issue.

The Modern View

Boards and CFOs now treat sustainability performance as a genuine input to risk pricing, access to capital, and long-term competitiveness — not a side project.

Five Categories of Sustainability Risk

Every major sustainability failure a company faces falls into one of these five buckets.



Regulatory Risk

Carbon pricing, mandatory disclosure (BRSR, CSRD), and outright bans on products or practices.



Physical Risk

Extreme weather damaging assets, disrupting supply chains, or straining water and resource access.



Reputational Risk

Consumer backlash, boycotts, and greenwashing accusations that damage brand trust.



Litigation Risk

Lawsuits from investors, regulators, or communities over emissions, pollution, or misleading claims.



Stranded Asset Risk

Assets that lose most of their value before the end of their useful life, due to policy or market shifts.

◆ INTERACTIVE

CASE STUDY

When Risk Actually Materializes: "Dieselgate"



Volkswagen Emissions Scandal (September 2015)

Volkswagen admitted to installing "defeat device" software in roughly 11 million diesel vehicles worldwide, designed to cheat regulatory emissions tests while polluting far more on real roads.

Financial Cost

More than $30 billion in fines, penalties, buybacks, and settlements globally — one of the largest corporate scandals in history.

Market Reaction

Volkswagen's share price dropped sharply within days of the disclosure, erasing billions in market value almost immediately.

Leadership Cost

The CEO resigned within days, and the company's reputation for engineering integrity took years to begin repairing.

The Lesson

This started as a regulatory compliance failure — but it became simultaneously a legal, financial, and reputational risk, all at once.

Stranded Assets: When Value Simply Disappears

The risk that is hardest for traditional financial models to see coming.

The Definition

An asset that suffers unanticipated or premature write-downs, devaluations, or conversion to liabilities — before the end of its expected economic life.

The Classic Example

Fossil fuel reserves currently on company balance sheets may never be extractable if climate policy tightens fast enough — reserves priced as assets today, worthless tomorrow.

Why It's Different From Other Risks

Stranded asset risk isn't about an accident or a scandal — it is the market itself re-pricing an entire asset class as the energy transition accelerates.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

Which category of sustainability risk best describes fossil fuel reserves that may become unextractable due to future climate policy?

A Reputational risk

B Litigation risk

C Stranded asset risk ✓

D Physical risk

ANSWER & WHY

C. Stranded asset risk

Stranded asset risk specifically refers to assets that lose most of their expected value before the end of their useful economic life — exactly the situation facing fossil fuel reserves as climate policy and the energy transition accelerate.

SECTION 2 OF 4

The Opportunity Landscape

Where sustainability stops being a defensive cost and becomes an actual source of competitive advantage.

Five Categories of Sustainability Opportunity

The flip side of the risk landscape — where genuine value gets created, not just protected.



Eco-Efficiency

Cutting energy, water, and material costs directly through better resource use — savings that hit the bottom line immediately.



Innovation & New Markets

Sustainability constraints often force genuinely new products, services, and business models into existence.



Brand & Customer Loyalty

A credible sustainability record increasingly shapes purchasing decisions, especially among younger consumers.



Access to Capital

Green finance and ESG-driven investment flows can materially lower the cost of raising capital.



Talent Attraction

Purpose-driven employer branding is now a real factor in recruiting and retaining skilled employees.

Porter's Hypothesis: Regulation Can Drive Innovation

A genuinely counterintuitive claim from one of the most cited strategists in business history.

The Conventional View	Environmental regulation is a cost that makes firms less competitive — compliance diverts resources away from productive investment.
Porter's Counter-Argument (Porter & van der Linde, 1995)	Well-designed regulation can trigger innovation that partially or fully offsets its own compliance cost — an "innovation offset."
The Mechanism	Regulation forces firms to notice resource inefficiencies they had been ignoring — cutting waste, energy, and materials is very often also cutting cost.

Creating Shared Value (Porter & Kramer, 2011)

A framework for finding competitive advantage inside social and environmental problems, not despite them.

Reconceiving Products & Markets	Designing products and services that address unmet social needs — turning underserved populations into new markets.
Redefining Productivity in the Value Chain	Rethinking energy use, logistics, sourcing, and labor practices as productivity levers, not just cost centers.
Building Supportive Industry Clusters	Investing in the local suppliers, infrastructure, and skills a company's own success actually depends on.

SECTION 2 · A CRITICAL LOOK

Shared Value Has Its Critics Too

Not every scholar accepted "Creating Shared Value" as the breakthrough Porter and Kramer claimed — the critique matters as much as the framework.

Not Entirely New

Critics (Crane et al., 2014) noted the concept closely echoed earlier work — Elkington's Triple Bottom Line, Prahalad and Hart's "base of the pyramid" thinking — without fully crediting it.

Ignores Real Trade-offs

The "win-win" framing can understate genuine tensions between profit and social good — not every social problem has a profitable solution waiting inside it.

A Shallow View of Responsibility

Focusing only on where social good and profit overlap can let companies ignore harms that fall outside that overlap entirely.

Still Widely Used, With Eyes Open

The framework remains genuinely useful for finding strategic opportunity — as long as it isn't mistaken for the whole of a company's ethical responsibility.

◆ INTERACTIVE

CASE STUDY

When Opportunity Was Seized Early: Ørsted

From "DONG Energy" to the World's Leading Offshore Wind Developer

A majority Danish state-owned oil and gas company made a deliberate strategic bet in the late 2000s to pivot toward offshore wind — divesting its oil and gas business and rebranding as Ørsted in 2017.

The Risk They Avoided

By exiting fossil fuels well ahead of the market, Ørsted sidestepped much of the stranded-asset exposure now facing slower-moving oil and gas majors.

The Opportunity They Captured

Early, sustained investment in offshore wind technology built engineering expertise and project experience competitors could not quickly replicate.

The Market Outcome

Ørsted became widely recognized as one of the world's leading offshore wind developers, turning a former fossil fuel utility into a renewable energy leader.

The Lesson

The company didn't just reduce its risk profile — it converted an anticipated regulatory and market shift into a durable competitive advantage.

◆ INTERACTIVE

CLASSIFY THE COMPANY ACTION

Risk Mitigation or Opportunity Capture?

For each real company action, decide whether it primarily defends against risk or actively captures new opportunity.

1. Diversifying suppliers away from a climate-exposed region
2. Launching a new product line for an underserved market
3. Purchasing insurance against extreme weather losses
4. Investing early in a technology competitors haven't adopted
5. Setting aside reserves for potential climate litigation
6. Redesigning a product line to open a new customer segment

RISK MITIGATION	OPPORTUNITY CAPTURE
1	2
3	4
5	6

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

What did Porter and van der Linde (1995) argue well-designed environmental regulation could trigger?

A Guaranteed bankruptcy for regulated firms

B An "innovation offset" that can partially or fully offset compliance costs ✓

C A permanent reduction in industry profitability

D No effect on firm behavior whatsoever

ANSWER & WHY

B. An "innovation offset" that can partially or fully offset compliance costs

Porter and van der Linde argued that well-designed regulation forces firms to notice inefficiencies they had been ignoring, triggering innovation that can partially or fully offset the cost of compliance — the foundational argument behind treating sustainability as a competitive opportunity.

SECTION 3 OF 4

Stakeholder Theory & Engagement

Who a company actually owes value to — and the frameworks for deciding what matters most.

Shareholder Primacy vs. Stakeholder Theory

Shareholder Primacy (Friedman, 1970)

"The social responsibility of business is to increase its profits" — famously argued in Milton Friedman's 1970 New York Times essay.

Managers are agents of shareholders; other obligations should be handled through law and market forces, not managerial discretion.

Simple, singular objective function: maximize shareholder returns within the law.

Stakeholder Theory (Freeman, 1984)

A business exists to create value for all groups with a stake in it — not shareholders alone (Freeman, "Strategic Management: A Stakeholder Approach").

Customers, employees, suppliers, communities, and shareholders are all legitimate stakeholders whose interests deserve consideration.

Requires balancing multiple, sometimes competing interests — genuinely harder to execute, and to measure.

SECTION 3 · A MODERN TURNING POINT

The Business Roundtable Statement (2019)

181 CEOs of America's largest companies publicly broke with over two decades of stated shareholder primacy.

> *"We commit to a fundamental commitment to all of our stakeholders — customers, employees, suppliers, communities, and shareholders." — Business Roundtable, August 19, 2019*

What Changed

Every Business Roundtable governance statement since 1997 had endorsed shareholder primacy — this reversed that stance explicitly.

The Critical Follow-Up

A widely cited study (Bebchuk & Tallarita, 2020–22) found most signatory companies made no changes to governance documents or executive pay structures afterward.

The "PR Hypothesis" vs. "Commitment Hypothesis"

Researchers frame this as an open empirical question: was the statement a genuine shift, or mainly a public-relations gesture?

The Stakeholder Salience Model

Not every stakeholder can be prioritized equally — this model gives managers a structured way to decide.

Power	Does this stakeholder have the ability to actually influence the organization — through markets, regulation, or public pressure?
Legitimacy	Is this stakeholder's claim on the organization socially or legally recognized as appropriate?
Urgency	How time-sensitive or critical is this stakeholder's claim right now?
The Framework (Mitchell, Agle & Wood, 1997)	Stakeholders holding all three attributes ("definitive stakeholders") demand the most immediate management attention — a practical triage tool.

Materiality Assessment

How a company actually decides which sustainability issues deserve its limited attention and resources.

The Core Question

Which issues are both significant to the business's financial performance and significant to its stakeholders — and where do the two overlap most?

The Materiality Matrix

A common visual tool plots issues on two axes — importance to stakeholders, and importance to business success — surfacing the handful that matter most.

Why It Matters Here

This is the practical bridge between stakeholder theory and corporate strategy — it turns "consider all stakeholders" into an actual, prioritized action list.

◆ INTERACTIVE

THINK & SHARE

"Using the Stakeholder Salience Model, who would you rank as the single most 'definitive' stakeholder for a large tech company today — and why?"

Consider power, legitimacy, and urgency together, not just one.

⏱ 3 MIN

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the Stakeholder Salience Model (Mitchell, Agle & Wood, 1997), what three attributes determine a stakeholder's priority?

A Size, revenue, and location

B Power, legitimacy, and urgency ✓

C Age, gender, and income

D Loyalty, tenure, and satisfaction

ANSWER & WHY

B. Power, legitimacy, and urgency

The Stakeholder Salience Model ranks stakeholders by power (ability to influence), legitimacy (socially/legally recognized claim), and urgency (time-sensitivity of the claim) — those holding all three are classified as "definitive" stakeholders demanding the most attention.

SECTION 4 OF 4

Long-Term Value Creation

How everything covered today ultimately shows up in the actual price of capital.

ESG Investing & Access to Capital

Sustainability performance increasingly determines not just reputation, but the actual cost of a company's money.

The Growth of ESG Investing

Environmental, Social, and Governance criteria are now a standard input for a large and growing share of global institutional investment decisions.

Why Investors Care

ESG performance is used as a proxy for management quality and long-term risk exposure — not (primarily) as an ethical statement by the investor.

The Practical Effect

Strong sustainability performers increasingly find it easier, and sometimes cheaper, to raise both debt and equity capital than weaker performers in the same industry.

SECTION 4 · WORKED EXAMPLE

How Sustainability Risk Prices Into the Cost of Capital

GIVEN

- Two companies each raise ₹100 crore via a 10-year corporate bond
- Company A (strong ESG rating): investors demand a 4.5% interest rate
- Company B (weak ESG rating, higher perceived climate/governance risk): investors demand 5.2%

FORMULA

Annual Interest Cost = Principal × Interest Rate

STEP-BY-STEP CALCULATION

1. Company A: ₹100 crore × 4.5% = ₹4.5 crore per year
2. Company B: ₹100 crore × 5.2% = ₹5.2 crore per year
3. Extra annual cost for Company B: ₹0.7 crore — roughly ₹7 crore over the life of the bond, purely from the market's perception of risk.

₹7 Cr Extra Cost

A weaker sustainability risk profile alone can cost a company crores over the life of a single bond — this is what "sustainability affects the cost of capital" means in practice.

Reporting Frameworks That Signal Long-Term Value

Investors and regulators increasingly expect structured, comparable sustainability disclosure — not a one-off marketing document.

GRI Standards

The most widely used global framework for sustainability reporting across environmental, social, and economic impact.

India's BRSR

SEBI's mandatory Business Responsibility and Sustainability Reporting for India's top 1,000 listed companies.

Integrated Reporting

Combines financial and sustainability performance into a single report, showing how they actually connect — not two separate documents.

Science Based Targets (SBTi)

Independently verifies that a company's emissions targets are actually consistent with the Paris Agreement's temperature goals.

◆ INTERACTIVE

CHECK YOUR UNDERSTANDING

In the cost-of-capital example, why did Company B pay a higher interest rate than Company A?

A Company B borrowed a larger amount of money

B Investors priced in greater perceived climate and governance risk ✓

C Company B had a shorter loan term

D Interest rates are unrelated to a company's sustainability profile

ANSWER & WHY

B. Investors priced in greater perceived climate and governance risk

Investors price bonds based on perceived risk. A weaker ESG rating signals greater exposure to regulatory, physical, and governance risk — so investors demand a higher interest rate to compensate for that additional risk, directly raising the company's cost of capital.

BRINGING IT TOGETHER

The Business Case, End to End

Four ideas, one continuous argument — from what can go wrong, to what a company is worth because of how it responds.

The Risk Landscape	Names the downside: regulatory, physical, reputational, litigation, and stranded-asset risk — real, costed, and no longer avoidable.
The Opportunity Landscape	Reframes the same terrain as competitive advantage — eco-efficiency, innovation, brand, capital, and talent.
Stakeholder Theory & Engagement	Supplies the "who": which stakeholders a company must actually satisfy, and how to prioritize among them.
Long-Term Value Creation	Shows where it all lands — in the market's own pricing of a company's capital, today.

◆ INTERACTIVE

APPLY IT ALL

"Pick one real company you know well. Where does it sit on the risk landscape, the opportunity landscape — and which stakeholder is it currently underweighting?"

This is the exact exercise a sustainability strategy team runs in practice.

⏱ 4 MIN